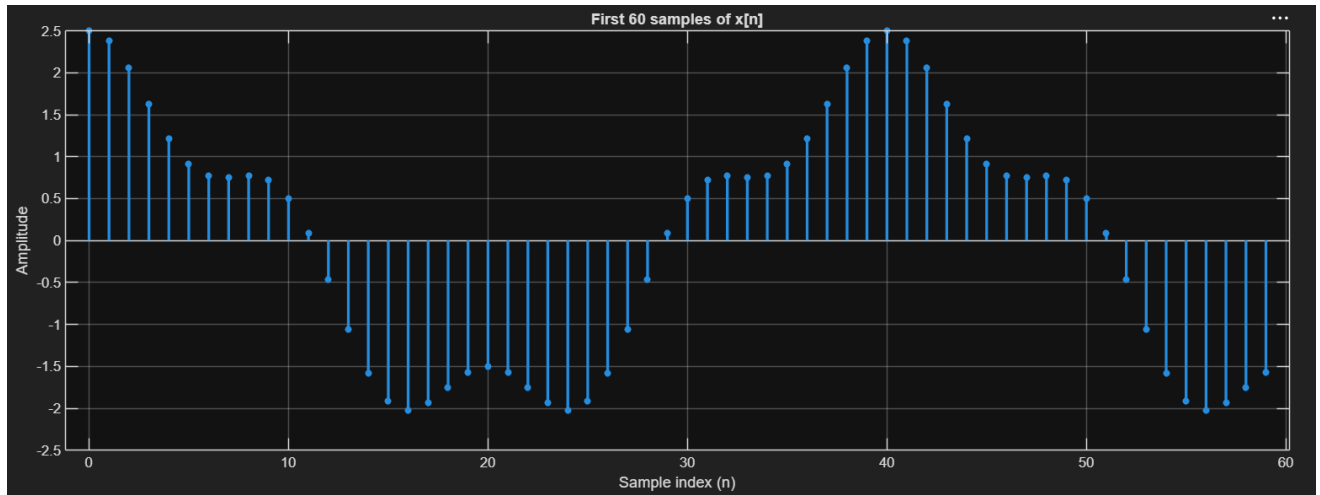
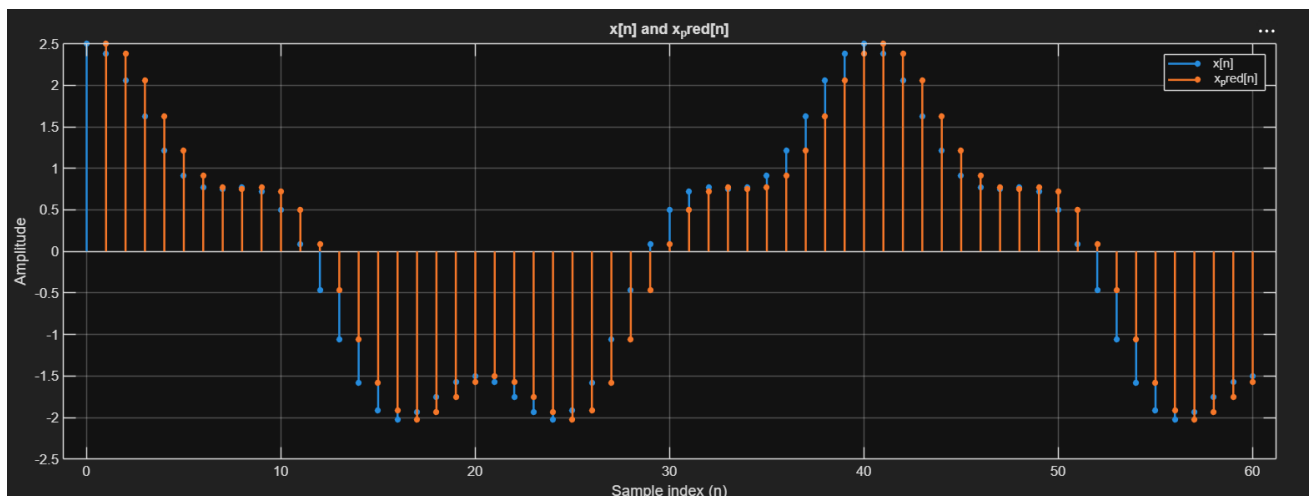


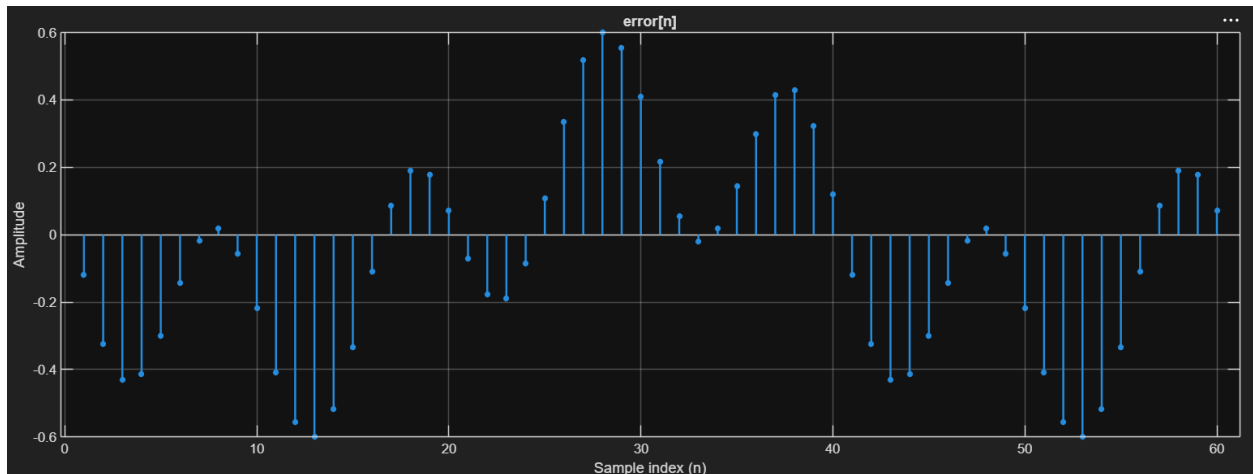
Part D – MATLAB Investigation

D1 – Generate and plot signals



D2 – Simple Predictor





Question: From the figures, explain why $e[n]$ is generally smaller than $x[n]$.

Answer: because the signal has significant correlation between adjacent samples, x and x_{pred} are close to each other, which reduces the difference between them, thus reducing the value of $e[n]$.

D3 – Prediction Gain

After calculating P_x , P_e , G_p , $G_p(\text{db})$ by MATLAB code we got these values:

```

46
47 %===== D3 =====
48
49 N = length(x);
50
51 Px = (1/N) * sum(abs(x).^2);
52 Pe = (1/(N - 1)) * sum(abs(e(2:end)).^2);
53
54 Gp = Px/Pe;
55
56 Gp_db = 10 * log10(Gp);
57
Command Window
Px = 2.125000
Pe = 0.097407
Gp = 21.815616
Gp(db) = 13.387675
>>

```

Question: Calculate the prediction gain. What does a large prediction gain mean?

Answer:

Prediction Gain turned out to be $21.81 = (13.39 \text{ db})$.

Prediction Gain means basically gives an idea of how much smaller the power of $e[n]$ is compared to $x[n]$, so a large G_p means that prediction error has significantly lower power than the original signal, indicating that the predictor is working well.